

Business Problem Definition Document

Project 11: Fake News & Misinformation Detection System

March 2025

1 Business context and motivation

The rapid spread of misinformation has become one of the most critical challenges facing modern society. With billions of users sharing content across social media platforms, news websites, and messaging applications, false information can spread faster than the truth causing real-world harm.

Industries affected

Industry	Impact
News and media	Credibility loss, audience distrust
Healthcare	Vaccine hesitancy, dangerous health advice
Finance	Stock manipulation via false rumors
Politics	Election interference, public opinion manipulation
E-commerce	Fake product reviews, brand damage

2 Target users and stakeholders

Stakeholder	Role	Need
Content Moderators	Primary users	Automated pre-screening to reduce workload
News Platforms	System deployers	Flag suspicious articles before publication
Journalists / Fact-checkers	Secondary users	Quick credibility assessment tool
General Public	End beneficiaries	Access to more reliable information
Social Media Companies	Business clients	At-scale content filtering
Government / Regulators	Oversight bodies	Compliance with misinformation laws

3 Problem being solved

3.1 Core problem statement

Manual fact-checking cannot scale to match the volume and velocity of online content. Human reviewers can check only a fraction of published articles, leaving the majority of misinformation undetected until after significant damage is done.

3.2 Specific pain points

- **Speed:** A human fact-checker takes about 15–30 minutes per article but content spreads in seconds.
- **Volume:** Platforms publish millions of articles/posts per day.
- **Cost:** Maintaining large moderation teams is expensive.
- **Consistency:** Human reviewers have subjective biases and inconsistent judgments.
- **Scale:** No single organization can monitor the entire information ecosystem.

3.3 What the system does

- Accept a news article, headline, or social media post as input.

- Classify it as **Real**, **Fake** or **Uncertain** (with confidence score).
- Provide an explanation of why it was flagged.
- Optionally trigger agentic fact-checking via external sources when confidence is low.

4 Why NLP is required

- **Linguistic patterns:** Fake news uses sensationalist language, emotional manipulation, and specific writing styles detectable by NLP.
- **Semantic understanding:** Simple keyword matching fails; context and meaning matter.
- **Entity recognition:** Identifying people, organizations, and locations helps verify claims.
- **Cross-document reasoning:** Comparing claims against known facts requires language understanding.
- **Scale requirement:** Only automated NLP can process millions of documents.

Why traditional approaches fail

- Rule-based systems are easily circumvented.
- Metadata-only approaches miss content-level signals.
- Image/video detection alone is insufficient.

5 Success Metrics

5.1 Technical Metrics

Metric	Target	Rationale
F1-Score	≥ 0.85	Balance between precision and recall
Precision	≥ 0.85	Minimize false positives (wrongly flagging real news)
Recall	≥ 0.80	Minimize false negatives (missing fake news)
AUC-ROC	≥ 0.90	Overall model discriminative ability
Inference Latency	$\leq 500\text{ ms}$	Real-time usability requirement
Accuracy	≥ 0.85	Overall classification correctness

5.2 Business Metrics

Metric	Target	How to Measure
Review time reduction	60% faster	Average time per article before vs after
Moderator workload reduction	Handle 5x more content	Articles processed per moderator/day
Cost per review	Reduced by 40%	Operational cost comparison
False positive rate	$< 5\%$	% of real news incorrectly flagged
User trust score	Increase 20%	Platform user surveys
Response time	< 1 minute for flagging	Time for submission to decision

5.3 Acceptance Thresholds

- **Minimum acceptable:** $F1 \geq 0.8$, Latency $\leq 1\text{ s}$
- **Production ready:** $F1 \geq 0.85$, Latency $\leq 500\text{ ms}$
- **Excellent:** $F1 \geq 0.9$, Latency $\leq 200\text{ ms}$

6 Scope and Constraints

6.1 In scope

- English-language news articles and headlines.

- Binary classification (Real vs Fake) + confidence score.
- REST API deployment.
- Agentic fact-checking for uncertain cases.

6.2 Out of scope

- Video/Image misinformation.
- Real-time social media stream processing.
- Non-English languages.
- Legal liability decisions.